```
clc;
clear;
close all;

%% ============================================================
% DESIGN AND COMPARATIVE ANALYSIS OF
% BUTTERWORTH, CHEBYSHEV-I AND ELLIPTIC IIR FILTERS
% ============================================================

%% 1. INPUT PARAMETERS
Fs = 1000;      % Sampling frequency (Hz)
Fp = 100;       % Passband edge (Hz)
Fst = 150;      % Stopband edge (Hz)
Rp = 1;         % Passband ripple (dB)
Rs = 40;        % Stopband attenuation (dB)

% Normalized frequencies
Wp = Fp/(Fs/2);
Wst = Fst/(Fs/2);

%% 2. CHEBYSHEV TYPE-I FORMULA
epsilon = sqrt(10^(Rp/10)-1);

N_formula = acosh(sqrt((10^(Rs/10)-1)/(10^(Rp/10)-1))) ...
            / acosh(Fst/Fp);

N_theory = ceil(N_formula);

%% 3. BUTTERWORTH FILTER
[Nb, Wb] = buttord(Wp, Wst, Rp, Rs);
[bB, aB] = butter(Nb, Wb, 'low');

%% 4. CHEBYSHEV TYPE-I FILTER
[Nc, Wc] = cheb1ord(Wp, Wst, Rp, Rs);
[bC, aC] = cheby1(Nc, Rp, Wc, 'low');

%% 5. ELLIPTIC FILTER
[Ne, We] = ellipord(Wp, Wst, Rp, Rs);
[bE, aE] = ellip(Ne, Rp, Rs, We, 'low');

%% 6. DISPLAY INPUT PARAMETERS
fprintf('\n=================================================\n');
```

```
=================================================
```

```
fprintf('       IIR FILTER COMPARATIVE ANALYSIS\n');
```

```
       IIR FILTER COMPARATIVE ANALYSIS
```

```
fprintf('=================================================\n');
```

```
==================================================
```

```
fprintf('\nINPUT PARAMETERS\n');
```

```
INPUT PARAMETERS
```

```
fprintf('Sampling Frequency = %.2f Hz\n', Fs);
```

```
Sampling Frequency = 1000.00 Hz
```

```
fprintf('Passband Edge Frequency = %.2f Hz\n', Fp);
```

```
Passband Edge Frequency = 100.00 Hz
```

```
fprintf('Stopband Edge Frequency = %.2f Hz\n', Fst);
```

```
Stopband Edge Frequency = 150.00 Hz
```

```
fprintf('Passband Ripple = %.2f dB\n', Rp);
```

```
Passband Ripple = 1.00 dB
```

```
fprintf('Stopband Attenuation = %.2f dB\n', Rs);
```

```
Stopband Attenuation = 40.00 dB
```

```
%% 7. CHEBYSHEV FORMULA RESULTS
fprintf('\nCHEBYSHEV TYPE-I FORMULA\n');
```

```
CHEBYSHEV TYPE-I FORMULA
```

```
fprintf('Ripple factor epsilon = %.4f\n', epsilon);
```

```
Ripple factor epsilon = 0.5088
```

```
fprintf('Calculated order = %.4f\n', N_formula);
```

```
Calculated order = 6.2071
```

```
fprintf('Theoretical minimum N = %d\n', N_theory);
```

```
Theoretical minimum N = 7
```

```
%% 8. FILTER ORDER COMPARISON
fprintf('\n==================================================\n');
```

```
==================================================
```

```
fprintf('             FILTER ORDER COMPARISON\n');
```

```
            FILTER ORDER COMPARISON
```

```
fprintf('==================================================\n');
```

```
================================================
```

```
fprintf('Butterworth Order = %d\n', Nb);
```

```
Butterworth Order = 12
```

```
fprintf('Chebyshev Type-I Order = %d\n', Nc);
```

```
Chebyshev Type-I Order = 6
```

```
fprintf('Elliptic Order = %d\n', Ne);
```

```
Elliptic Order = 4
```

```
%% 9. DISPLAY FILTER COEFFICIENTS
fprintf('\n================================================\n');
```

```
================================================
```

```
fprintf('          BUTTERWORTH COEFFICIENTS\n');
```

```
          BUTTERWORTH COEFFICIENTS
```

```
fprintf('================================================\n');
```

```
================================================
```

```
fprintf('Numerator b =\n');
```

```
Numerator b =
```

```
disp(bB);
```

```
   1.0e-03 *

    0.0002    0.0027    0.0146    0.0488    0.1098    0.1757    0.2050    0.1757    0.1098    0.0488    0.
```

```
fprintf('Denominator a =\n');
```

```
Denominator a =
```

```
disp(aB);
```

```
    1.0000   -6.8844   22.4341  -45.5100   63.7725  -64.8501   48.9643  -27.6092   11.5218   -3.4662    0.
```

```
fprintf('\n================================================\n');
```

```
================================================
```

```
fprintf('        CHEBYSHEV TYPE-I COEFFICIENTS\n');
```

```
        CHEBYSHEV TYPE-I COEFFICIENTS
```

```
fprintf('=====================================================\n');
```

```
=====================================================
```

```
fprintf('Numerator b =\n');
```

```
Numerator b =
```

```
disp(bC);
```

```
   1.0e-03 *

    0.0464    0.2782    0.6956    0.9274    0.6956    0.2782    0.0464
```

```
fprintf('Denominator a =\n');
```

```
Denominator a =
```

```
disp(aC);
```

```
    1.0000   -4.8694   10.3807  -12.3367    8.5969   -3.3268    0.5586
```

```
fprintf('\n=====================================================\n');
```

```
=====================================================
```

```
fprintf('              ELLIPTIC COEFFICIENTS\n');
```

```
             ELLIPTIC COEFFICIENTS
```

```
fprintf('=====================================================\n');
```

```
=====================================================
```

```
fprintf('Numerator b =\n');
```

```
Numerator b =
```

```
disp(bE);
```

```
    0.0197   -0.0171    0.0333   -0.0171    0.0197
```

```
fprintf('Denominator a =\n');
```

```
Denominator a =
```

```
disp(aE);
```

```
    1.0000   -3.0330    3.8118   -2.2911    0.5554
```

```
%% 10. FREQUENCY RESPONSES
N = 4096;
```

```
[HB, f] = freqz(bB, aB, N, Fs);
[HC, ~] = freqz(bC, aC, N, Fs);
[HE, ~] = freqz(bE, aE, N, Fs);

MB = 20*log10(abs(HB) + eps);
MC = 20*log10(abs(HC) + eps);
ME = 20*log10(abs(HE) + eps);

%% 11. PASSBAND RIPPLE
P = f <= Fp;

Ripple_B = max(MB(P)) - min(MB(P));
Ripple_C = max(MC(P)) - min(MC(P));
Ripple_E = max(ME(P)) - min(ME(P));

%% 12. STOPBAND ATTENUATION
S = f >= Fst;

Stop_B = -max(MB(S));
Stop_C = -max(MC(S));
Stop_E = -max(ME(S));

%% 13. PERCENTAGE DEVIATION
Dev_C_Ripple = abs(Ripple_C - Rp)/Rp*100;

Dev_B_Stop = abs(Stop_B - Rs)/Rs*100;
Dev_C_Stop = abs(Stop_C - Rs)/Rs*100;
Dev_E_Stop = abs(Stop_E - Rs)/Rs*100;

%% 14. TRANSITION CHARACTERISTICS
TransitionBW = Fst - Fp;

%% 15. DISPLAY PERFORMANCE
fprintf('\n=====================================================\n');
```

```
=====================================================
```

```
fprintf('                FILTER PERFORMANCE\n');
```

```
               FILTER PERFORMANCE
```

```
fprintf('=====================================================\n');
```

```
=====================================================
```

```
fprintf('\nPASSBAND RIPPLE\n');
```

```
PASSBAND RIPPLE
```

```
fprintf('Butterworth = %.4f dB\n', Ripple_B);
```

```
Butterworth = 0.8033 dB
```

```
fprintf('Chebyshev Type-I = %.4f dB\n', Ripple_C);
```

```
Chebyshev Type-I = 1.0000 dB
```

```
fprintf('Elliptic = %.4f dB\n', Ripple_E);
```

```
Elliptic = 1.0000 dB
```

```
fprintf('\nSTOPBAND ATTENUATION\n');
```

```
STOPBAND ATTENUATION
```

```
fprintf('Butterworth = %.4f dB\n', Stop_B);
```

```
Butterworth = 40.0198 dB
```

```
fprintf('Chebyshev Type-I = %.4f dB\n', Stop_C);
```

```
Chebyshev Type-I = 41.3365 dB
```

```
fprintf('Elliptic = %.4f dB\n', Stop_E);
```

```
Elliptic = 40.0000 dB
```

```
fprintf('\nCHEBYSHEV RIPPLE DEVIATION\n');
```

```
CHEBYSHEV RIPPLE DEVIATION
```

```
fprintf('Specified Ripple = %.2f dB\n', Rp);
```

```
Specified Ripple = 1.00 dB
```

```
fprintf('Measured Ripple = %.4f dB\n', Ripple_C);
```

```
Measured Ripple = 1.0000 dB
```

```
fprintf('Percentage Deviation = %.4f %%\n', Dev_C_Ripple);
```

```
Percentage Deviation = 0.0000 %
```

```
fprintf('\nSTOPBAND ATTENUATION DEVIATION\n');
```

```
STOPBAND ATTENUATION DEVIATION
```

```
fprintf('Butterworth Deviation = %.4f %%\n', Dev_B_Stop);
```

```
Butterworth Deviation = 0.0494 %
```

```
fprintf('Chebyshev-I Deviation = %.4f %%\n', Dev_C_Stop);
```

```
Chebyshev-I Deviation = 3.3413 %
```

```
fprintf('Elliptic Deviation = %.4f %%\n', Dev_E_Stop);
```

```
Elliptic Deviation = 0.0000 %
```

```
fprintf('\nTRANSITION CHARACTERISTICS\n');
```

```
TRANSITION CHARACTERISTICS
```

```
fprintf('Passband Edge = %.2f Hz\n', Fp);
```

```
Passband Edge = 100.00 Hz
```

```
fprintf('Stopband Edge = %.2f Hz\n', Fst);
```

```
Stopband Edge = 150.00 Hz
```

```
fprintf('Transition Band = %.2f Hz\n', TransitionBW);
```

```
Transition Band = 50.00 Hz
```

```
%% 16. MAGNITUDE RESPONSE
figure;

plot(f, MB, 'LineWidth', 1.5);
hold on;
plot(f, MC, 'LineWidth', 1.5);
plot(f, ME, 'LineWidth', 1.5);

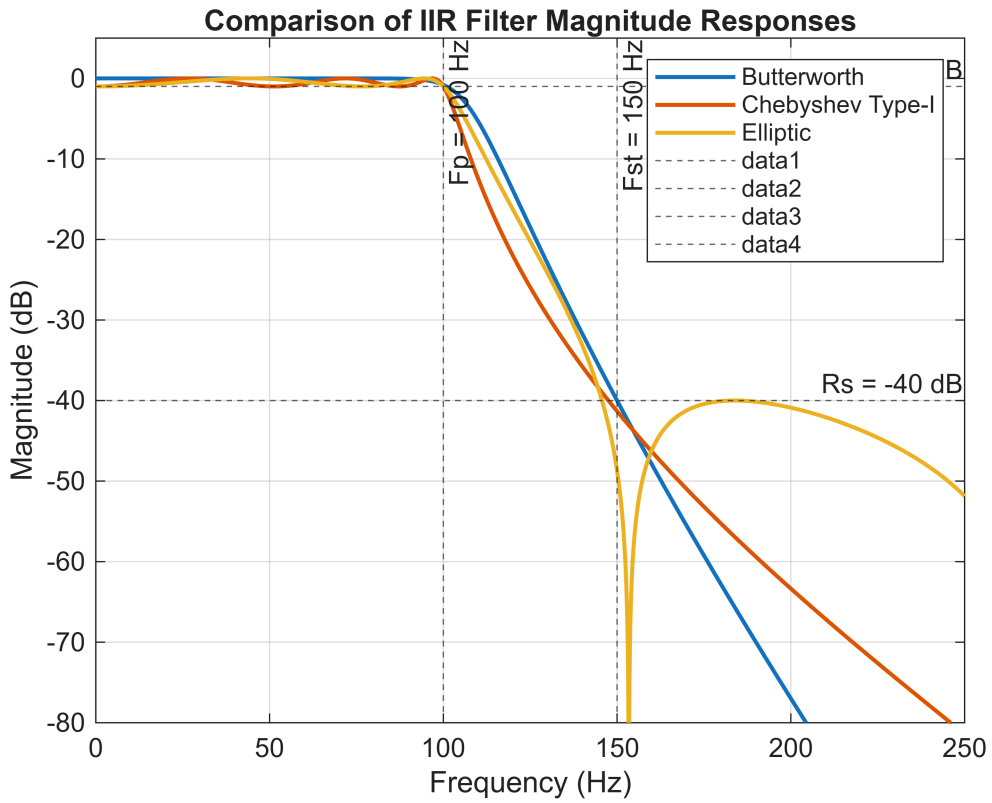
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Comparison of IIR Filter Magnitude Responses');

legend('Butterworth', 'Chebyshev Type-I', 'Elliptic');

xlim([0 250]);
ylim([-80 5]);

xline(Fp, '--', 'Fp = 100 Hz');
xline(Fst, '--', 'Fst = 150 Hz');
yline(-Rp, '--', 'Rp = -1 dB');
yline(-Rs, '--', 'Rs = -40 dB');

hold off;
```




```
%% 17. PHASE RESPONSE
Phase_B = unwrap(angle(HB))*180/pi;
Phase_C = unwrap(angle(HC))*180/pi;
Phase_E = unwrap(angle(HE))*180/pi;

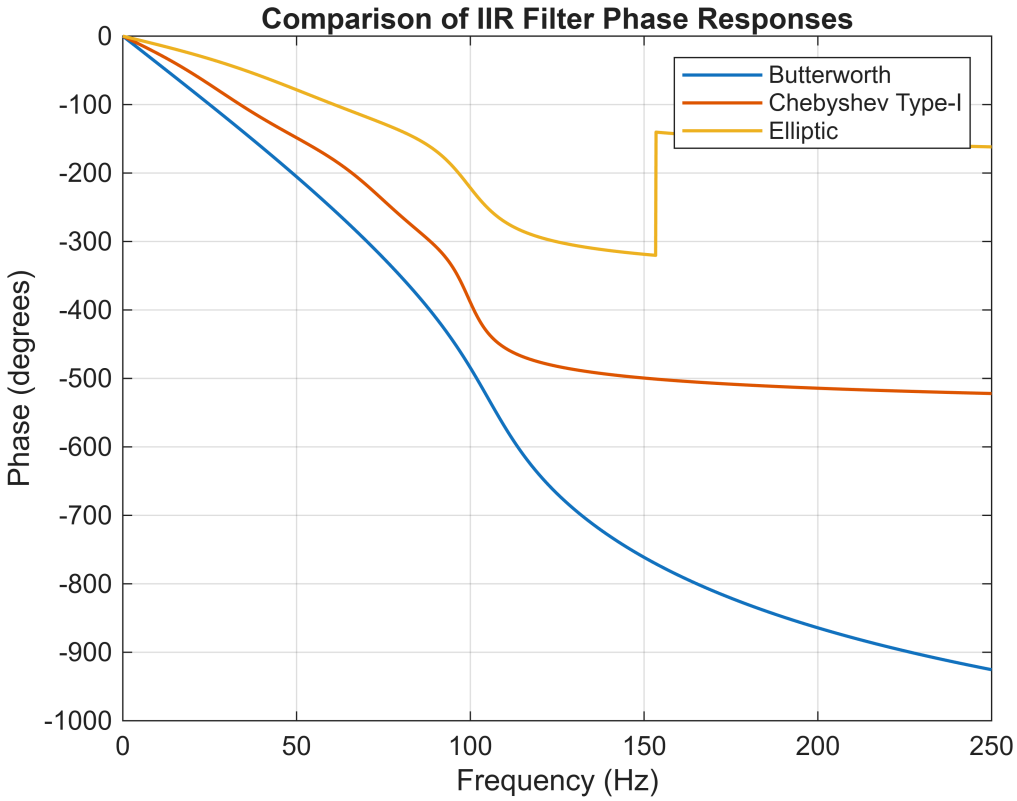
figure;

plot(f, Phase_B, 'LineWidth', 1.2);
hold on;
plot(f, Phase_C, 'LineWidth', 1.2);
plot(f, Phase_E, 'LineWidth', 1.2);

grid on;
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
title('Comparison of IIR Filter Phase Responses');

legend('Butterworth', 'Chebyshev Type-I', 'Elliptic');

xlim([0 250]);
hold off;
```



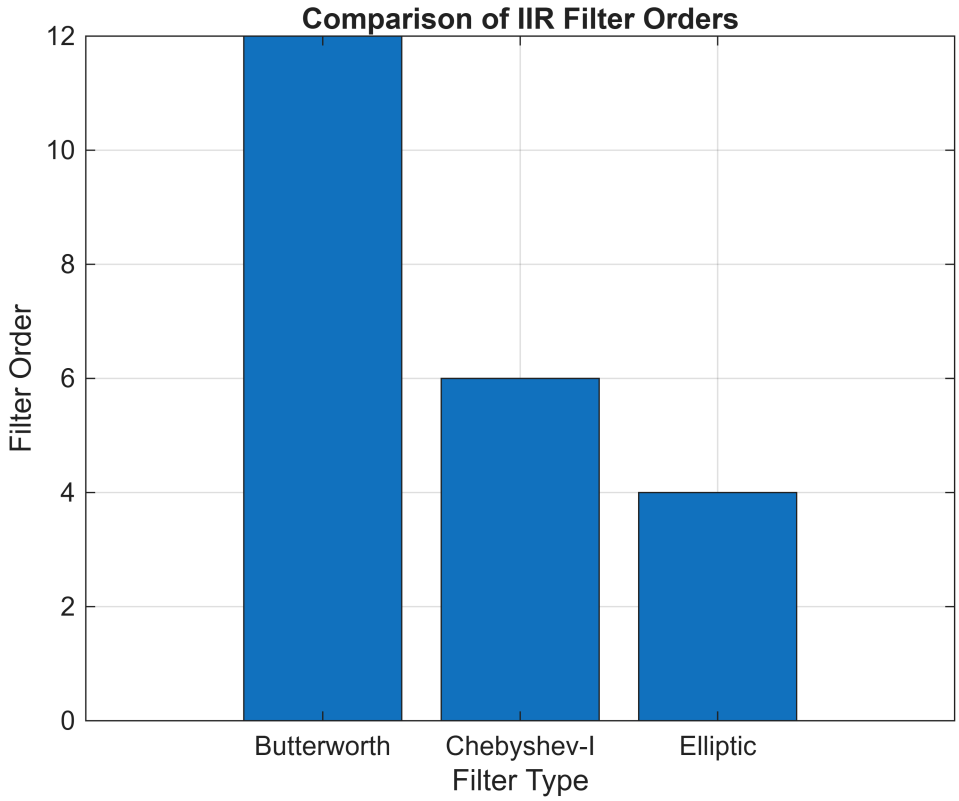

```
%% 18. FILTER ORDER GRAPH
figure;

bar([Nb Nc Ne]);

set(gca, 'XTick', 1:3);
set(gca, 'XTickLabel', ...
    {'Butterworth', 'Chebyshev-I', 'Elliptic'});

ylabel('Filter Order');
xlabel('Filter Type');
title('Comparison of IIR Filter Orders');

grid on;
```


Comparison of IIR Filter Orders
12
10
8
6
4
2
0
Filter Order
Butterworth
Chebyshev-I
Elliptic
Filter Type